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The Men and Women of Project Firestar—How the United States Developed the Labor Force to Colonize Titan

A Backcasting Analysis of Labor Force Requirements for U.S. Technoeconomic Leadership Through 2050

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About This Paper

In this paper, we examine the changes necessary to produce a U.S. labor force sufficient to sustain technoeconomic leadership through 2050 while achieving a prestige technoeconomic marvel comparable to that achieved with the successful manned mission to the moon in 1969. We use a backcasting approach as a rhetorical tool; the fictional author is a RAND analyst writing in the year 2050 and reflecting on both (1) the technical marvel that the United States was able to achieve and (2) the changes in the U.S. labor force that occurred during the 2025–2050 time frame that made this possible. However, like all RAND analysis, this paper is objective and evidence-based, and all the insights and implicit policy recommendations are based on data and findings available at the time that this paper was completed (October 2025).

The views, opinions, and/or findings expressed are those of the authors and should not be interpreted as representing the official views or policies of DARPA or the U.S. government.

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Summary

A significant expansion in the U.S. technical labor force is likely to be necessary to sustain U.S. technoeconomic leadership through 2050. In this paper, we conduct rough order-of-magnitude estimates of workforce needs in a plausible scenario requiring accelerated U.S. technological innovation and deployment. A rapid shift in workforce capacity demands would likely pose a challenge based on current U.S. approaches to job matching, higher-level education, and on-the-job training; we introduce and estimate the cost and impact of policy solutions to optimize and supplement the U.S. workforce to achieve continuing technological advantage.

Key Takeaways

Our analysis has the following key observations:

- A *tripling* of doctorate-level science, technology, engineering, and mathematics (STEM) researchers by 2050—from 900,000 (in 2025) to 3 million (in 2050)—will likely be necessary to sustain U.S. technoeconomic leadership while achieving a prestige technological marvel.
- Although the United States could potentially double the size of its STEM research workforce—from 900,000 to 1.75 million—international collaborations that allow the United States to access research talent among allies and partners will be necessary to sustain technoeconomic leadership.
- A nearly 70-percent increase in the number of U.S. engineers and technicians graduating each year—from 145,000 (in 2025) to 240,000 (during 2030–2040)—will be equally critical for sustained competitiveness, although the United States should be able to produce these individuals itself.
- Building the U.S. technical labor force is projected to cost approximately $1.1 trillion over ten years, although more than half of this cost could be reasonably offset by U.S. private sector investments, including in-kind support.

Although labor-augmenting technologies, such as artificial intelligence (AI), may offset some of the necessary expansions, these technologies are becoming increasingly accessible by U.S. competitors and are unlikely to give the United States a competitive edge. China, in particular, offers a near-term threat to U.S. technoeconomic leadership given its recent and ongoing advances in AI and high-performance computing. This analysis aims to sketch a way forward for the United States to maintain its technoeconomic dominance.

Contents

The Men and Women of Project Firestar—How the United States Developed the Labor Force to Colonize Titan: A Backcasting Analysis of Labor Force Requirements for U.S. Technoeconomic Leadership Through 2050

This paper is written from the perspective of a RAND analyst in the year 2050, employing a "backcasting" analysis that details a specific desired end state and then describes what was required to achieve that end state.[1] *The scenario presents an ambitious goal selected for its relevance to plausible U.S. objectives and requirements for the accelerated development of existing technologies. This analysis—building on existing (as of 2025) analyses of U.S. workforce needs, resources, and costs—provides rough order-of-magnitude estimates of the labor force capacity necessary to sustain technological leadership and how the United States might meet that target. In light of future uncertainty, our policy design does not rely on perfect accuracy of estimates herein.*

During the past two decades, the United States has reestablished its global technoeconomic leadership. Although the People's Republic of China (PRC) beat the United States to Mars with a manned mission in 2030, the U.S. colony on the Saturnian moon of Titan—now entering its third year of autonomous operation as we enter the second half of the 21st century—proved that no technoeconomic feat is too large for the collective intellectual and engineering might of the United States.

The colonization of Titan was made possible by a series of technological and engineering breakthroughs over the past 20 years, making what was only theoretically feasible in 2025 a reality today. These breakthroughs were accelerated by early-generation artificial intelligence (AI) capabilities developed in the 2030s, but it was U.S. scientists, engineers, and technicians—and the vibrant private

[1] For a discussion of the backcasting approach in a workshop context, see Robert J. Lempert, Steven W. Popper, Endy Y. Min, and James A. Dewar, eds., *Shaping Tomorrow Today: Near-Term Steps Towards Long-Term Goals*, RAND Corporation, CF-267-RPC, 2009, p. xii.

sector that was willing to take necessary risks when underwritten by the U.S. government—that made this feat possible.

This retrospective paper aims to describe how the United States built this labor force. The colonization of Titan would not have been possible without those who have become its public face—specifically, the astronauts who volunteered for and the trillionaires who financed key elements of this one-way mission. However, countless articles and books have already been written about these men and women, and their stories need not be told again. Our focus here is instead to describe the human capital on which these achievements depended; we focus on the scientists, engineers, and technicians necessary to develop, produce, test, evaluate, and deploy the required technologies at the necessary scale.

Our analysis fills a gap in the existing literature, as there is no single accounting of the labor force that made this possible. This accounting is complicated by the fact that the research community pursued dozens of technological pathways simultaneously—as it was unclear from the onset which concepts would prove viable—and thousands of private companies were involved. However, we can derive back-of-the-envelope estimates of the total labor force that was deployed based on a diagnostic of the variety of technologies pursued and the magnitude of public and private investments involved.

At its peak, Project Firestar[2]—which became in 2035 the household term for the set of U.S. government projects that made the Titan mission possible—involved more than 300,000 dedicated doctorate-level researchers across a variety of fields of science, technology, engineering, and mathematics (STEM) to drive the necessary scientific advances and 450,000 engineers and technicians to build, test, validate, and deploy these technologies. Alongside these focused efforts, the United States needed to train an additional 1.8 million researchers and 500,000 engineers and technicians to sustain U.S. technological advantage across every domain important for U.S. economic competitiveness and national security, for a total workforce requirement of more than 3 million scientists, engineers, and technicians. This combined expansion of the scientific and technical workforce has been central to sustained U.S. technoeconomic leadership.

Project Firestar was nearly one order of magnitude larger than Project Apollo, its U.S. historical antecedent that landed the first human on the moon in 1969, in terms of total manpower.[3] However, like Project Apollo,[4] Project Firestar has developed the capacity for U.S.-led innovation in nearly every sector relevant for today's ongoing technoeconomic competition.

This paper—after describing the magnitude of the achievement that was accomplished—describes the public-private collaboration of the National Defense Talent Act of 2030 (NDTA) that made this possible. This collaboration allowed for more than $1.1 trillion in new investments in U.S. technology workers over ten years, consisting of $320 billion in new federal funding for workforce development

[2] This name is from Michael Flynn's *Firestar*, which offers an inspired focus on the importance of holistic education reform as a necessary ingredient in a successful U.S. space program (Michael Flynn, *Firestar: An Epic of Science Fiction for the Twenty-First Century*, Tor Books, 1996).

[3] At the height of Project Apollo, the National Aeronautics and Space Administration (NASA) employed a total of 411,000 U.S. government civilian employees and contractors (see Table 3-26 in Jane Van Nimmen, Leonard C. Bruno, and Robert L. Rosholt, *NASA Historical Data Book, 1958–1968*: Vol. 1, *NASA Resources*, National Aeronautics and Space Administration, NASA SP-4012, 1976).

[4] Loura Hall, "Going to the Moon Was Hard—But the Benefits Were Huge, for All of Us," National Aeronautics and Space Administration, July 15, 2019.

and a roughly $800 billion private investment in workforce training over ten years.[5] This massive private investment was driven by a change in the U.S. tax code that—borrowing a tactic used previously by the PRC[6]—allowed private employers participating in Project Firestar to deduct 200 percent of all investments in training and workforce development.

Project Firestar

In 2028, the PRC achieved a strategic surprise—in an achievement reminiscent of the Soviet Union's 1957 Sputnik launch—by repurposing its planned Tianwen-3 from a sample return mission to a predeployment mission for an eventual human colony on Mars.[7] Following the Mars One playbook, the PRC launched its first crewed mission in 2030 with an initial four taikonauts—which landed successfully on the Red Planet in 2031—and has added four more taikonauts every two years since.[8] Although the PRC Mars colony has remained heavily dependent on resupply from Earth and has become increasingly expensive for the PRC to sustain, it initially served as a marker of Chinese leadership in space.[9]

Facing a loss of international prestige, the U.S. government launched Project Firestar and committed the United States to establish what was then described as the "first self-sustaining human colony" by 2050. Rather than Mars, the United States turned its focus to Titan, "where conditions are right for a self-sustaining, long-term human settlement" for a very simple reason: Titan has an atmosphere.[10] The manned mission launched from low earth orbit in 2046 with no potential resupply for several years.[11]

The success of the Titan mission was made possible by rapid advancements in propulsion, manufacturing, and food production technologies. These advancements—which occurred alongside other enabling technologies (e.g., life support systems) that were somewhat easier to achieve[12]—allowed the United States to overcome three of the core challenges facing a Titan mission. The three challenges faced in 2025 can be summarized as follows:

- **Propulsion.** Ensuring the physical and mental health of the astronauts required a total one-directional travel time of 18 months.[13] In 2025, a six-year journey was achievable with existing

[5] All dollar values for estimated costs are inflation-adjusted to 2025 dollars using the Consumer Price Index.

[6] Alex Muresianu, "U.S. Must Fix R&D Treatment to Compete with China," *Tax Policy Blog*, Tax Foundation, March 6, 2024.

[7] Tianwen-3 was, as of 2022, speculated to be the first PRC unmanned return mission to Mars (Andrew Jones, "Tianwen-3: China's Mars Sample Return Mission," The Planetary Society, September 15, 2022).

[8] Sydney Do, Andrew Owens, Koki Ho, Samuel Schreiner, and Olivier de Weck, "An Independent Assessment of the Technical Feasibility of the Mars One Mission Plan—Updated Analysis," *Acta Astronautica*, Vol. 120, March–April 2016.

[9] Thomas Corbett and Peter W. Singer, "China's Plan to Rule the Heavens," *Defense One*, January 17, 2025.

[10] Amanda Hendrix, "Confession of a Planetary Scientist: 'I Do Not Want to Live on Mars,'" NPR, October 16, 2017.

[11] The launch date is based on the data for a hypothetical Titan mission used in Marco Gajeri, Paolo Aime, and Roman Ya. Kezerashvili, "A Titan Mission Using the Direct Fusion Drive," *Acta Astronautica*, Vol. 180, March 2021.

[12] See discussion in Volker Maiwald, Mika Bauerfeind, Svenja Fälker, Bjarne Westphal, and Christian Bach, "About Feasibility of SpaceX's Human Exploration Mars Mission Scenario with Starship," *Scientific Reports*, May 23, 2024.

[13] The specific number of 18 months and the idea of learning from the PRC mission is from Charles Wohlforth and Amanda R. Hendrix, *Beyond Earth: Our Path to a New Home in the Planets*, Pantheon, 2016.

chemical propulsion technology,[14] a 3.5-year journey was possible with experimental nuclear rocket engines,[15] and a two-year travel time was theoretically possible with experimental fusion drives.[16]

- **Manufacturing.** Minimizing mission payload while maximizing flexibility in colony design required manufacturing approaches that could satisfy all fabrication and adjacent requirements using locally abundant hydrocarbons.[17] In 2025, the bioprinting of human tissue and food was only theoretically feasible, and existing platforms lacked the precision and flexibility for complex outputs or the bootstrapping necessary to build increasingly large-scale features (such as insulated greenhouses).[18]
- **Food.** The long-term viability of a Titan colony depended on secure and stable access to nutritious food. While there has always been sufficient atmosphere, water, and light on Titan to support agriculture[19]—and greenhouses developed for Mars were able to be manufactured in situ on Titan using available technology—synthetic biology was only just beginning to develop approaches to address the challenges that Titan's microgravity (which is comparable to that of Earth's moon) creates for plants.[20]

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[14] The Hohmann transfer orbit from Earth orbit to the surface of Titan requires a total change in velocity (delta-V) of 11,430 m/s and approximately six years (as compared with 5,710 m/s and eight months for Mars), assuming aerobraking for purposes of deceleration (Ulysse Carion, "The Solar System: A Subway Map," infographic, undated). The delta-V available to the SpaceX Starship in 2019 was a reported 6,500 m/s (Jeff Becker, "A Starcruiser for Space Force: Thinking Through the Imminent Transformation of Spacepower," *War on the Rocks*, May 19, 2021). The Starship will have to achieve a delta-V of at least 8,000 m/s for the return trip from the Earth's orbit to the moon and back again (Carion, undated), which is currently scheduled for 2026.

[15] The Defense Advanced Research Projects Agency's (DARPA's) Demonstration Rocket for Agile Cislunar Operations has a theoretical *specific impulse*—a measure of an engine's efficiency in using reaction mass calculated as the ratio of change in momentum (impulse) per mass of propellant—of 900 seconds, which is more than double that of SpaceX's Raptor, which has a proven impulse of about 380 seconds (Jon Kelvey, "NASA and DARPA Are Cautioned Against Overselling the Performance of Their Nuclear Rocket Tech," Aerospace America, February 1, 2023). One estimate (for the Mars mission) suggests that this approximate doubling of specific impulse could reduce travel time by 40 percent—from 260 days to 150 days (A. McKelvy, "This is a great question, but the other answer negates your premise straight away," answer to Kasie Ream, "What is the relationship between delta v and the time taken to reach a destination?" Stack Exchange, March 21, 2025).

[16] Experimental fusion drives as of 2025 can currently generate thrust of at least 100 newtons (Pulsar Fusion, "Sunbird," webpage, undated). This would enable a mission time to Titan of two years for 12,500 kg of payload under constant acceleration (estimated based on analysis presented in Gajeri, Aime, and Kezerashvili, 2021). Configurations of multiple engines could allow for missions with larger payloads, such as the proposed configuration of six such engines that would allow the same mission time for a payload of 75,000 kg (S. A. Cohen, C. Swanson, N. Mcgreivy, A. Raja, E. Evans, P. Jandovitz, M. Khodak, G. Pajerj, T. D. Rognuen, S. Thomas, and M. Paluszek, "Direct Fusion Drive for Interstellar Exploration," *Journal of the British Interplanetary Society*, Vol. 72, No. 2, February 2019).

[17] See, for example, Wohlforth and Hendrix, 2016, p. 254.

[18] Longfei Zhou, Jenna Miller, Jeremiah Vezza, Maksim Mayster, Muhammad Raffay, Quentin Justice, Zainab Al Tamimi, Gavyn Hansotte, Lavanya Devi Sunkara, and Jessica Bernat, "Additive Manufacturing: A Comprehensive Review," *Sensors*, Vol. 24, No. 9, May 2024.

[19] Christopher McKay concludes that "[l]ight levels on the surface of Titan are more than adequate for photosynthesis" despite the fact that "the maximum level of sunlight on the surface of Titan is about 0.1% that of the overhead sun on Earth's surface" (Christopher P. McKay, "Titan as the Abode of Life," *Life*, Vol. 6, No. 1, March 2016).

[20] Lindsey Berebitsky, "Purdue and NASA Scientists Plant the Seeds for Lunar Agriculture," Purdue University, September 10, 2024.

Labor Requirements for Research and Development

The U.S. scientific workforce expanded dramatically in the wake of the Tianwen-3 launch. In addition to focused investment in the three technological domains that were understood at the time to be core impediments to the success of the Titan mission, a simultaneous concerted effort was aimed to ensure that the United States did not fall behind in other technological domains.

Nearly 300,000 individual doctorate-level researchers were focused on developing the technologies necessary for the Titan mission at the peak of research and development (R&D) efforts in 2045, with each of the three technological domains requiring a level of effort comparable to that of Project Apollo.[21] These researchers worked to rapidly accelerate the basic science underlying each of the core domains, as well as to deploy and validate these capabilities.[22] Alongside these Titan-focused efforts, a scientific research workforce of nearly 2.7 million researchers—approximately triple the size of this workforce in 2025—was necessary to ensure sustained U.S. technological leadership across all domains critical to U.S. economic security.[23]

This threefold growth—from a total of just over 900,000 doctorate-level researchers in 2025 to nearly 3 million by 2045—represented a massive expansion of the scientific workforce.[24] Initially, the authors of the NDTA envisioned that the new public and private financing would make it possible to meet this required tripling in the number of researchers with only U.S. citizens.

However, the NDTA's assessment tool (discussed in a subsequent section) rapidly made it clear that there were simply not enough U.S. citizens with the proficiencies necessary for success in these roles. Out of necessity and industry pressure, lawmakers amended the NDTA to support

[21] The peak number of scientists and engineers working on Project Apollo was 91,700 in 1966; the 300,000 reported in the text is a tripling of that number. The estimate from Project Apollo includes both U.S. government civilians and contractors working on the effort (see Table 3-27 in Van Nimmen, Bruno, and Rosholt, 1976).

[22] NASA's occupational codes were not sufficient at the time to distinguish between doctorate-level researchers and others, although both engineers and scientists were equally critical to the scientific discoveries of the time (see, for example, Sylvia Doughty Fries, *NASA Engineers and the Age of Apollo*, National Aeronautics and Space Administration, NASA SP-4104, 1992).

[23] This is our estimate of the number of researchers that would be necessary to sustain a constant growth rate in U.S. total factor productivity. This estimate is based on a 2020 journal article in which the researchers demonstrate—using sectoral analysis (e.g., agriculture, semiconductors), firm-level analysis, and analysis of the census of manufacturing—that U.S. average researcher productivity falls by at least 4.0 percent per year (Nicholas Bloom, Charles I. Jones, John Van Reenen, and Michael Webb, "Are Ideas Getting Harder to Find?" *American Economic Review*, Vol. 110, No. 4, April 2020; see Figure 1 and Table A1, which offers the authors' more conservative estimate of decreasing productivity). This rate of productivity reduction suggests that a scientific workforce of constant size would experience an approximate 64-percent reduction in productivity over 25 years and that a tripling of the overall scientific research workforce—from approximately 900,000 in 2025 to 2.7 million in 2050—would be necessary independent of any other types of idea-augmenting technological change (e.g., AI). Other researchers report comparable estimates of decreasing researcher productivity, including some who report a 50-percent reduction in R&D productivity over a decade (Fabio Pammolli, Laura Magazzini, and Massimo Riccaboni, "The Productivity Crisis in Pharmaceutical R&D," *Nature Reviews Drug Discovery*, June 2011, p. 430), and some who report an 80- to 90-percent reduction in disruptiveness of published research during the 1980–2010 time frame (Michael Park, Erin Leahey, and Russell J. Funk, "Papers and Patents Are Becoming Less Disruptive over Time," *Nature*, January 4, 2023, p. 139); both suggest that our estimates using the findings from Bloom et al. (2020) represent a conservative estimate.

[24] There were just over 900,000 U.S.-trained doctorate-level researchers in science, engineering, or health fields working in the United States in 2023; this included approximately 800,000 working full-time and 100,000 working part-time (National Center for Science and Engineer Statistics, "Table 1-1: U.S. Residing Doctoral Scientists and Engineers, by Fine Field of Doctorate and Employment Status: 2023," in *Survey of Doctorate Recipients: 2023*, NSF 25-321, January 23, 2025).

collaboration with U.S. allies and partners—which had long been a hallmark of U.S. technological development and deployment.

Labor Requirements for Development, Production, and Operations

At its peak in 2040, 450,000 engineers and technicians were involved in producing, testing, evaluating, and deploying the technologies necessary to land (e.g., propulsion) and sustain (e.g., manufacturing, food) a human mission on Titan.[25] This capacity was in addition to the scientists working to make the mission technologically possible and included a wide variety of technical capabilities, including aerospace engineers, electricians, mechanical engineers, plumbers, software engineers, toolmakers, and welders.[26]

In addition to the requirements of the Titan mission, a further 500,000 trained engineers and technicians were needed to meet the broader requirements of the U.S. economy as it adapted to ensure U.S. technoeconomic advantage. This represented an increase in the annual throughput of engineers and technicians of 50,000 each year from 2030 through 2040.[27] Thus, in combination with the Titan

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[25] This estimate is based on three assumptions:

- The total cost of the Titan mission is roughly double the cost of a mission to Mars, which was estimated to be approximately "half a trillion dollars" in 2014 (Harry W. Jones, "Humans to Mars Will Cost About 'Half a Trillion Dollars' and Life Support Roughly Two Billion Dollars," *46th International Conference on Environmental Systems*, July 10, 2016).
- The capital intensity of the advanced manufacturing technologies necessary for the Titan mission is comparable to that of the semiconductor industry (which is approximately 50 percent), suggesting total new investment of approximately $700 billion ("The Approaching Semiconductor Capital Expenditure Supercycle," *Silicon Matter*, Substack, April 3, 2024).
- Analysis from the advanced semiconductor industry indicates that $250 billion in new investment created 160,000 new job openings for engineers and technicians (Bill Wiseman, Brendan Jay, Nicholas Liao, Taylor Roundtree, and Wade Toller, "Reimagining Labor to Close the Expanding US Semiconductor Talent Gap," McKinsey & Company, August 2, 2024).

Using the fact that $1 in 2014 is approximately equal to $1.4 in 2025, we get our estimate: ($500 billion × 2 × 1.4) × 0.5 × (160,000 jobs) / ($250 billion) = 450,000 jobs.

[26] This job list is based on a nonrandom sample of job listings on SpaceX's website in June 2025 (SpaceX, "Open Positions," webpage, undated).

[27] This estimate of 500,000 is based on the annual difference—multiplied by ten to capture the total production over the decade between 2030 and 2040—between (1) the number of new job openings each year for engineers and technicians and (2) the annual production of engineers and technicians. For the first, we used data from the Bureau of Labor Statistics, which reported that "[a]bout 195,000 openings are projected each year, on average, in these occupations due to employment growth and the need to replace workers who leave the occupations permanently" (U.S. Bureau of Labor Statistics, "Architecture and Engineering Occupations," webpage, *Occupational Outlook Handbook: Architecture and Engineering Occupations*, last updated April 18, 2025a). For the second, we calculated the current production of U.S. engineers and technicians as follows:

- **Technicians (~55,000 per year).** This was calculated as the sum of the number of graduates from an associate's degree program with an engineering (~5,900) or engineering technology (~50,200) degree (Table 321.10 in National Center for Education Statistics, "List of 2023 Digest Tables," webpage, *Digest of Education Statistics*, 2023).
- **Engineers (~90,000 per year).** This was calculated based on the number of engineering students graduating each year from an undergraduate or master's program, adjusted by the number that transition into scientific fields and that are U.S. citizens: (1) In 2022, U.S. universities produced a total of approximately 166,000 engineering students—123,000 at the undergraduate level and 43,000 at the master's level (approximations based on Tables 322.10 and 323.10 in National Center for Education Statistics, 2023); (2) data from 2019 suggest that only 80 percent of the undergraduate

mission, a total of 950,000 new engineers and technicians needed to be trained over a decade. Building this workforce required a substantial increase in the production of U.S.-trained engineers and technicians. In the years before Project Firestar, only 145,000 U.S. engineers and technicians were entering the labor force annually.[28] Expanding this labor force by 950,000 over a decade required the United States to expand by roughly 65 percent—from 145,000 to 240,000—the annual production of engineers and technicians.[29]

National Defense Talent Act of 2030

To build the necessary human capital for Project Firestar, the U.S. President signed into law the NDTA on September 16, 2030. This act authorized $320 billion in direct public funding over a decade—$32 billion per year—for workforce training, STEM education, and labor market efficiency.[30] The act also created what have been described as "super deductions" for eligible private sector investments in training, which—at a cost of roughly $200 billion in tax incentives over ten years[31]—unlocked an estimated $800 billion in private investments to support labor force development over a decade.[32] Alongside these two labor force programs, the NDTA made possible a U.S.-hosted, private sector–led platform for research collaboration that allowed the United States to access the research talent it needed for technoeconomic leadership.

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students and 85 percent of master's level–students transition into scientific fields (Carolyn Wilson, *Current Status of the U.S. Engineering and Computing Workforce, 2019*, American Society for Engineering Education, 2019); and (3) one-third of engineers trained in the United States are foreign born (National Science Board, *The State of U.S. Science and Engineering 2024*, NSB-2024-3, March 2024). Our estimate follows as $(123{,}000 \times 0.8 + 43{,}000 \times 0.85) \times 2/3 \approx 90{,}000$.

The estimate of 500,000 reported in the text is thus calculated as $(195{,}000 - 55{,}000 - 90{,}000) \times 10 = 500{,}000$.

28 See previous footnote.

29 The 240,000 is calculated as 145,000 + (950,000/10 years).

30 The magnitude of this hypothetical public funding roughly equals the 2025 Workforce Innovation and Opportunity Act budget ($12 billion per year), plus additional financial assistance for higher education (roughly $20 billion per year). This estimate is sensitive to assumptions about individual funding and private sector engagement (see next section), and it is conditional on the continuation of existing financial aid and workforce training. Although we estimate that doubling support for STEM fields would directionally support the policy objectives, further research is needed to understand a more precise price of degree-switching among undergraduates and post–bachelor's degree workers (Benjamin Collins and David H. Bradley, *The Workforce Innovation and Opportunity Act and the One-Stop Delivery System*, Congressional Research Service, R44252, updated September 26, 2022; Federal Student Aid, "About Us," webpage, U.S. Department of Education, undated).

31 This calculation takes into account current tax deductions for education and training—which already allow employers to deduct ordinary and necessary training costs—and assumes an overall 100-percent increase in deductions over the total cost estimate for training ($800 billion). At a corporate tax rate of 21 percent, the cost is thus estimated at 21 percent of $800 billion, or $168 billion. Even if employers were to pay for training to qualify new employees, it is not clear that the estimated additional tax expenditure would amount to the full 200 percent (i.e., $336 billion) because of existing individual and business tax credits for education. For more on this topic, see Erica York, "Tax Treatment of Worker Training," Tax Foundation, March 21, 2019.

32 The magnitude of this hypothetical private sector contribution is perhaps on the conservative side; Apple alone is reported to have invested $55 billion per year in China, with much of that flowing into building a labor force there (Patrick McGee, *Apple in China: The Capture of the World's Greatest Company*, Scribner, 2025).

Component 1: Talent Assessment and Publicly Funded Education Subsidies

The NDTA's new approach to talent assessment—the American Advanced Aptitude Algorithm, or "Quad-A"—provided a much more efficient and effective mechanism for targeting financial and mentoring support to individuals with the aptitude and personality necessary for success in STEM careers. Before Quad-A's launch, the United States had only crude measures of potential for success, leaving many individuals without clarity on how they could best contribute their skills to the economy.[33] Quad-A overcame the numerous challenges with existing aptitude tests and provided a framework to enable efficient and effective targeting of merit aid and other support (such as mentoring).[34] The resulting job-matching improvements helped to expand talent pipelines, optimize training for individuals' strengths, and increase retention in otherwise hard-to-fill jobs.

This assessment tool enabled an increase in the number of students who completed doctorate-level degrees and stayed in STEM fields by more than 850,000 over ten years.[35] This ten-year program more than tripled the throughput of doctorate-level STEM researchers, which hovered at around 30,000 to 35,000 per year in the 2020s.[36]

Quad-A identified more than 1 million additional individuals who were inclined toward either engineering or technician roles.[37] Although these individuals were offered the same financial package as doctorate-level researchers, with a stipend covering up to five years of education and training in either a postsecondary institution or an internship program, the program experienced the same

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[33] Bryan Hancock and Brooke Weddle, "Right Skills, Right Person, Right Role," *McKinsey Talks Talent* podcast, 2023.

[34] Challenges with assessment approaches include test anxiety, testing scope, and "teaching to the test" (see, for example, Allison Morris, *Student Standardised Testing: Current Practices in OECD Countries and a Literature Review*, Organisation for Economic Co-operation and Development, OECD Education Working Paper No. 65, 2011).

[35] This estimate is based on four underlying assumptions:

- The average number of high school graduates during 2030–2040 is approximately 3.6 million (Western Interstate Commission of Higher Education, *Knocking at the College Door: Projections of High School Graduates*, December 2024).
- Approximately 11 percent of U.S. high school students were assessed as "top performers" in science in 2022, which provides a lower bar for the number of students likely to be tracked by the Quad-A, as they received a Level 5 or Level 6 score on the Organisation for Economic Co-operation and Development's (OECD's) Programme for International Student Assessment (OECD, "Table I.B1.3.3: Percentage of Students at Each Proficiency Level in Science," PISA 2022 dataset, 2022).
- Research suggests that at least 23 percent of students could be convinced to pursue a STEM career trajectory (Teng Zhao and Lara Perez-Falkner, "Perceived Abilities or Academic Interests? Longitudinal High School Science and Mathematics Effects on Postsecondary STEM Outcomes by Gender and Race," *International Journal of STEM Education*, June 22, 2022).
- The number of students that would have pursued a relevant doctorate degree without the intervention would have remained at approximately 32,000 per year as in 2022 (National Center for Science and Engineer Statistics, "Science, Technology, Engineering, and Mathematics (STEM) Education, by Gender," webpage, undated).

These assumptions produce the following estimate: (3.6 million students × 11% top performers – 32,000 already Ph.D. bound) × 23% could be incentivized.

[36] National Center for Science and Engineer Statistics, undated.

[37] This estimate follows a similar logic as for doctoral students but is instead focused on the 19.9 percent of students that were scored as "Level 4" in the sciences in the OECD's Programme for International Student Assessment (OECD, 2022). Using the data on the estimated number of engineers (90,000) and technicians (55,000) from the previous section, this results in the following estimate: (3.6 million students × 19.9% Level 4 – 145,000 already in sciences) × 10 years × 23% could be incentivized ≈ 1,300,000 students over ten years.

attrition—of about 20 percent—into business and other financial sectors that had prevailed in the years before the NDTA.[38] Still, this program provided the foundation for the rapid increase in the production of U.S. engineers and technicians that the United States required.

Component 2: Private Sector Support for Workforce Training

The targeted NDTA educational subsidies covered only part of the total cost of building the labor force that the United States needed for leadership in technoeconomic competition. Indeed, estimates from 2030 suggested that additional investments of $300,000 for technicians,[39] $500,000 for engineers,[40] and $550,000 for scientists were necessary to provide these individuals the skills and training that they needed to be useful.[41] While Congress initially considered financing these costs with public spending, the potential near-doubling of the costs of the program—the total cost of this

[38] The American Society for Engineering Education reports that "20% of bachelor's graduates, 14% of master's graduates, and 14% of doctoral graduates in engineering worked within a non-science or engineering related field for their principal job" in 2017 (Wilson, 2019).

[39] This estimate is the total cost of a four-year apprenticeship, including an aggregate administrative cost of $6,000 (Robert Lerman, Jessica Shakesprere, Daniel Kuehn, and Batia Katz, *What Are the Costs of Generating Apprenticeships? Findings from the American Apprenticeship Initiative Evaluation*, Abt Associates and Urban Institute, September 2022) and four years of salary for an entry-level high-skilled technician estimated at $70,000, which is estimated using three-fourths of the 2024 median wage of an electrical power line installer (U.S. Bureau of Labor Statistics, "Electrical Power-Line Installers and Repairers," webpage, *Occupational Outlook Handbook*, last updated April 18, 2025b). The sum of these two different elements is what is reported in the text: ($6,000 for administration) + (4 years of apprenticeship × $70,000 per year) ≈ $300,000.

[40] This estimate is based on five underlying assumptions:

- The average cost of a one-year master's degree is $60,000 (Melanie Hanson, "Average Cost of a Master's Degree: 2024 Analysis," webpage, Education Data Initiative, last updated November 23, 2024).
- The average timeline to qualification is four years (National Society of Professional Engineers, "Demonstrating Qualifying Engineering Experience for Licensure," webpage, July 28, 2007).
- Trainees receive a salary of $100,000 per year during their traineeship (U.S. Bureau of Labor Statistics, "17-2141 Mechanical Engineers," webpage, *Occupational Employment and Wage Statistics, May 2023*, last updated April 3, 2024).
- Each individual receives two paid summer internships valued at $12,000 each.
- Individuals receive monthly mentoring sessions over a five-year period valued at $200 per month.

The sum of these five different elements is what is reported in the text: ($60,000 for a master's degree) + (4 years of traineeship × $100,000 per year) + (2 summer internships × $12,000/each) + ($12,000 mentoring) ≈ $500,000.

[41] This estimate assumes a $100,000-per-year fellowship over three years—building on Google's existing program, which provides $85,000 per year (Google Research, "Google PhD Fellowship Program," webpage, undated)—with average compensation for early career Ph.D. graduates over two years ($120,000 per year) as reported in the National Science Foundation's 2019 Survey of Doctorate Recipients (Jean Opsomer, Angela Chen, Wan-Ying Chang, and Daniel Foley, *U.S. Employment Higher in the Private Sector Than in the Education Sector for U.S.-Trained Doctoral Scientists and Engineers: Findings from the 2019 Survey of Doctorate Recipients*, National Center for Science and Engineering Statistics, NSF 21-319, April 2021). Monthly mentoring sessions for five years, estimated at a value of $200 per session, are also included ($12,000 total). The sum of these three different elements is what is reported in the text: (3 years of fellowship × $100,000 per year) + (2 years of salary × $120,000/year) + ($12,000 mentoring) ≈ $550,000.

additional training was around $800 billion over ten years[42]—threatened to torpedo support for the bill among fiscal conservatives.[43]

Although Congress balked at underwriting the entire cost of developing this workforce, the private sector grew increasingly concerned that existing incentives were failing to produce enough graduates. Thus, a public-private deal was struck in which the nation's private employers would pay the costs of this additional training.[44] Private employers benefited in two ways. First, while investments in worker training were already considered a business expense and fully tax-deductible, the NDTA allowed employers participating in the program to deduct 200 percent of all training investments.[45] Second, these employers were provided early access to new talent.

This proposal dramatically scaled existing employer-provided fellowships and training models. More than 35,000 companies (dubbed "Patriot Employers") committed to fund fellowships and provide mentors and internship opportunities for newly minted doctorates, engineers, and technicians. The NDTA accounted for the largest year-over-year increase in employer-provided training on record. Subsequent findings of improved workforce productivity relative to nonparticipating firms made the case for continued employer engagement, laying the groundwork for the demand-led system of today.

Component 3: Private Sector–Led International Research Coordination

By 2032, it was clear that the NDTA would be unable to produce sufficient research talent in the United States to meet the objectives of Project Firestar. Although some early projections suggested that the incentives of the NDTA might produce as many as 225,000 additional doctorate-trajectory STEM students each year,[46] the reality was much lower, at 85,000. The consequence was that the United States would not have been able to build the scientific workforce that it needed—which required a 2.1-million-person expansion by 2045[47]—by itself.

As a result, entrepreneurial U.S. leaders initiated what is now widely described as "IMEC 2.0" as an international, U.S.-hosted, private sector–led platform for research collaboration with talent from

[42] This estimate is derived from the following calculation: (200,000 technicians × $300,000/technician) + (250,000 engineers × $500,000/engineer) + (850,000 doctoral researchers × $550,000/researcher).

[43] The Pew Research Center conducted a survey of U.S. adults in July 2021 on the topic of free college, finding that 75 percent of conservative Republicans opposed tuition-free college for students (Hannah Hartig, "Democrats Overwhelmingly Favor Free College Tuition, While Republicans Are Divided by Age, Education," Pew Research Center, August 11, 2021).

[44] See, for example, existing fellowship programs from companies, such as Google, that include both fellowships and mentoring (Google Research, undated).

[45] Daniel Bunn, "Tax Subsidies for R&D Spending and Patent Boxes in OECD Countries," Tax Foundation, March 17, 2021.

[46] This estimate is based on the upper end of what Zhao and Perez-Falkner (2022) suggested as the maximal number of students that might be shifted toward a STEM trajectory; they reported that as much as 62 percent of students (rather than 23 percent) could be persuaded if they were incentivized and convinced that they were high performers. Using the same logic as the estimate reported under Component 1, this estimate can be calculated as follows: (3.6 million students × 11% top performers − 32,000 already Ph.D. bound) × 62% could be incentivized ≈ 85,000 students per year.

[47] See discussion in the "Labor Requirements for Research and Development" section of this report.

around the world.[48] IMEC 2.0 is similar to its Belgian eponymous predecessor—the Interuniversity Microelectronics Centre, acronymized in lowercased letters as imec—in its ability to effectively manage intellectual property and thus enable public-private collaboration in leading-edge research with commercial applications.[49]

At the peak of IMEC 2.0, more than 500,000 international researchers from 90 countries participated in the U.S.-hosted center, making it roughly 100 times larger than its Belgian antecedent. This effort modified the regional innovation zones that had been included in the initial NDTA—by explicitly requiring international collaboration to receive the NDTA subsidies—as an additional tool to rapidly incubate and commercialize cutting-edge technologies.[50] The roughly 40 tech hubs associated with IMEC 2.0 would generate some $50 billion per year in annual economic output and involve the participation of roughly 500,000 researchers and engineers from across the world.[51]

Looking Ahead: U.S. Labor Force Requirements for Strategic Competition from 2050 Through 2075

Anticipating future U.S. labor force requirements is inherently an uncertain science, and this uncertainty is amplified by the combination of emergent technologies and shifting strategic competitors in 2050. These technologies include the about-to-be-deployed GPT-Z—which experts believe that China will be unable to effectively compete against and that is anticipated to fundamentally reshape U.S. research and development—and efforts by U.S. competitors, particularly the PRC, to leverage advances in synthetic biology to increase cognitive performance. However, the biggest wildcard is the rise of India, which has seen a *quintupling* of its relative share in the global economy over the past 25 years, since 2025.[52]

Although these uncertainties should not be overlooked, three core observations from this historical analysis summarize what we have learned from the past 25 years. The first is that substantial government investment in STEM education was critical to sustain U.S. technoeconomic leadership;

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[48] The idea of IMEC 2.0 was suggested in many discussions conducted during other components of RAND's support to DARPA for this project. However, others have similarly made the argument for a U.S. version of imec (see, for example, Arrian Ebrahimi and Jordan Schneider, "How to Make the NSTC a Moonshot Success," Institute for Progress, April 22, 2024).

[49] For a discussion of the imec intellectual property model, see Bart Leten, Wim Vanhaverbeke, Nadine Roijakkers, André Clerix, and Johan Van Helleputte, "IP Models to Orchestrate Innovation Ecosystems: IMEC, A Public Research Institute in Nano-Electronics," *California Management Review*, Vol. 55, No. 4, July 2013.

[50] This was a concept proposed by David Lin in hearings held by the U.S.-China Economic and Security Review Commission in February 2025 (U.S.-China Economic and Security Review Commission, "Hearing on 'Made in China 2025: Who Is Winning?'" February 6, 2025).

[51] Existing tech hubs in the United States are estimated to generate $1 billion to $2 billion in a single year (Cameron Davis, Ben Safran, Rachel Schaff, and Lauren Yayboke, "Building Innovation Ecosystems: Accelerating Tech Hub Growth," McKinsey & Company, February 28, 2023). The estimate of the number of IMEC 2.0 researchers is based on the size of the North Carolina Research Triangle regional innovation zone, which employs 13,000 individuals in advanced manufacturing (Work in the Triangle, "Advanced Manufacturing: Building Greatness in the Triangle," webpage, undated).

[52] As of 2025, India accounts for just 3.4 percent of global gross domestic product despite representing nearly one-fifth of the world's population. A variety of estimates suggest that a quintupling of its relative share in the global economy might be possible (D. K. Srivastava, "Indian Economy by 2050: In Pursuit to Achieve the $30 Trillion Mark," EY, August 25, 2022; Goldman Sachs, "How India Could Rise to the World's Second-Biggest Economy," July 6, 2023).

this was as true during U.S. competition with the Soviet Union in the 20th century as it has been these past 25 years. The second is that the private sector played a pivotal role in this competition, as both a source of funds and a coordinator. And the third is that the United States could not have achieved this alone; its population of qualified talent was simply too small to compete against its much more populous competitors. The secret weapon of the United States was its ability to seek and maintain partnerships with allies and partners around the globe.

Abbreviations

AI	artificial intelligence
DARPA	Defense Advanced Research Projects Agency
IMEC	Interuniversity Microelectronics Centre
NASA	National Aeronautics and Space Administration
NDTA	National Defense Talent Act of 2030
OECD	Organisation for Economic Co-operation and Development
PRC	People's Republic of China
R&D	research and development
STEM	science, technology, engineering, and mathematics

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